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FCT SERVICE NOTE SN – 06051009A

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Date: June 10, 2006

Subject: VCT Detector Plenum updates

Effectivity: VCT (7.x systems)

Details:

The detector plenum has been modified for improved reliability and functionality. The plenum has been stiffened to decrease flexing during rotation, connections have been reworked due to reliability issues and cable design/routing/mounting has been improved. The new plenum parts are not backward compatible with the current Plenum assembly. Older style plenum FRU's will only be available until the current stock is gone. When the old parts are gone they will change up to a plenum upgrade kit that will update your system to the new style plenum assembly. From that point on, your system will be able to use the new FRU's.

- New Plenum FRU's are NOT compatible with the original plenum.
- Original Plenum FRU's are NOT compatible with the new plenum.
- New systems will be shipped with the new style plenum.

The upgrade kit primary hardware is shown in Figure 1 (not shown are miscellaneous screws, etc). All the connections that are touched for most all service actions around the plenum have been moved to the front (no hidden connections for plenum removal/install). The only connection on the back is the RS232 communication between fan and heater controllers that is only touched when replacing either controller. The fan connections have changed to a molded sub-D connector for improved reliability over the molex connectors. The harness between the heater controller and routing boards has been significantly improved as well.

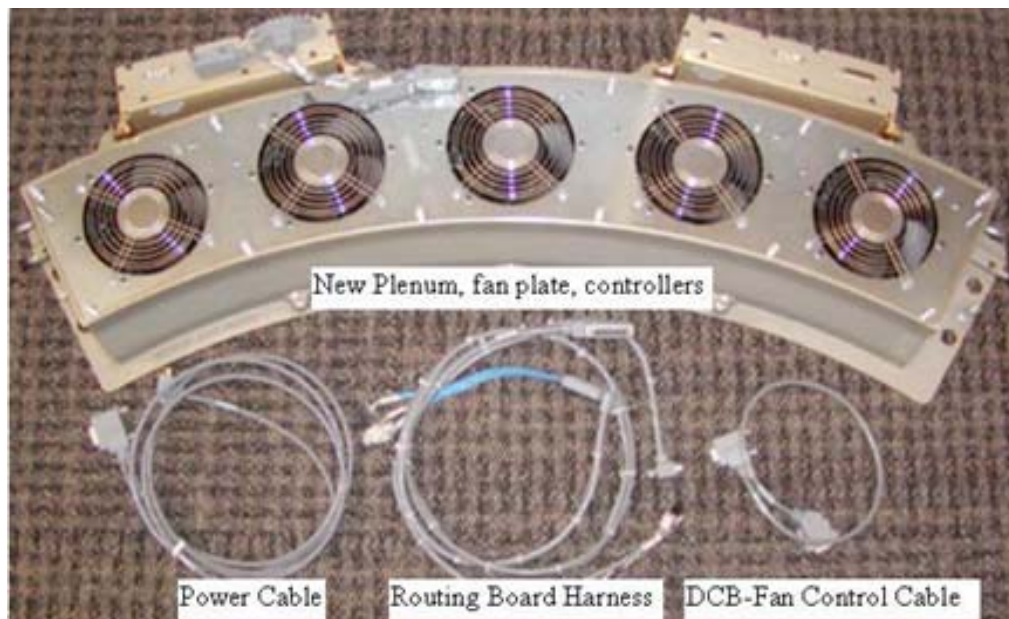


Figure 1 New Plenum Upgrade Kit

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For additional detail, reference 7.x service document replacement procedure: "Plenum Upgrade Procedure". Visually the style of plenum you have on the system is identified by the cable connections. Figure 2 shows the original style plenum. Note the cabling differences on the fan and heater controllers.

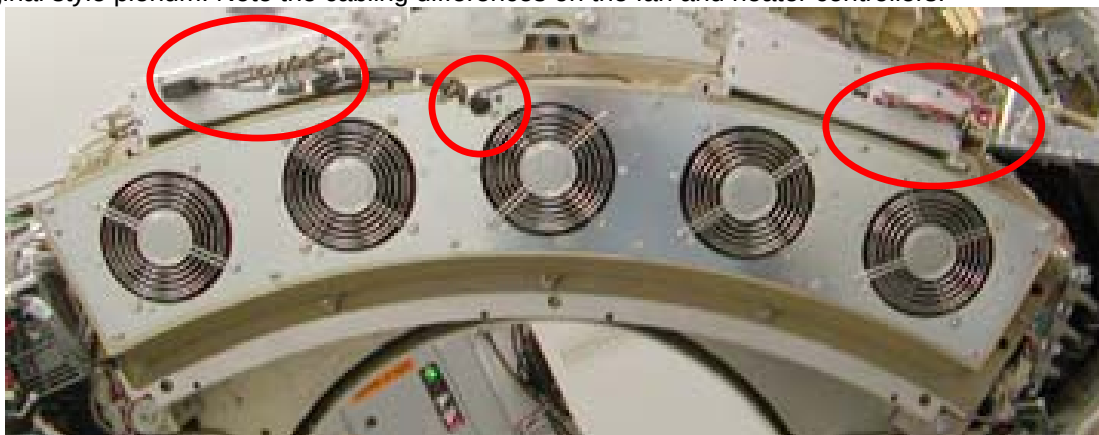


Figure 2 Original style Plenum (Visual differences highlighted)

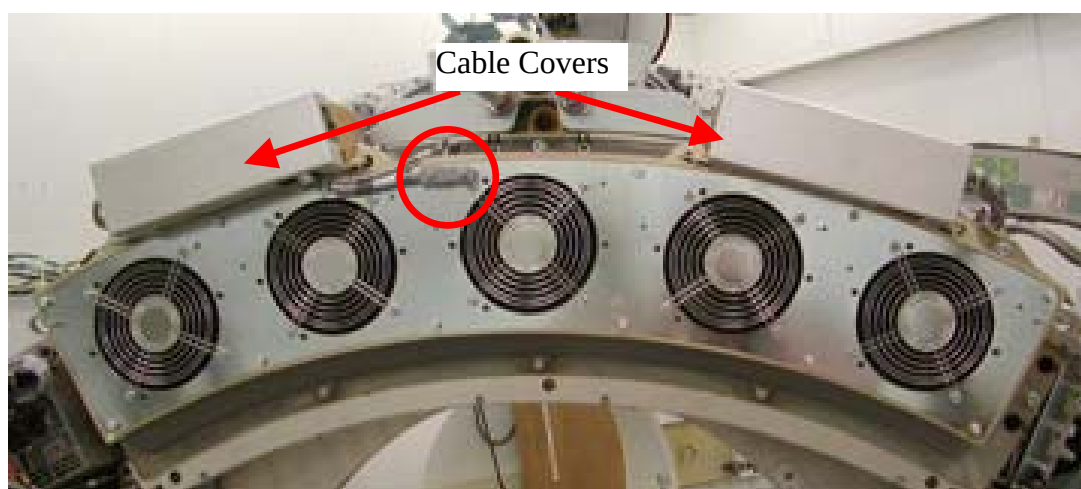


Figure 3 New Plenum with safety covers (Visual differences highlighted)

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Table 1 shows a listing of the original plenum part numbers and the new plenum part numbers for reference.

Table 1 Plenum FRU list

Description	Original FRU Part #	New FRU Part #
Plenum assembly	5117776	5169298
Routing board harness	5124291	5174115
DCB to CFC cable	5129701	5149165
48V power cable to controllers	5118036	5150272
Rotating CFC Assembly	5114020	5150071
DHC Assembly	5121621	5150070
Fan Plate Assembly	5120610	5158919
Plenum Filter Assembly	5113867	5168649
Plenum cable covers	NA	5170168
Plenum upgrade Kit	NA	5169770

If you have the original style plenum, continue to order parts for that plenum. If the old style part is no longer available the new upgrade kit will be sent instead (Part will change up automatically). The 7.x service instructions contain the procedure for performing the plenum upgrade under replacement parts. That procedure should be referenced if you receive a plenum upgrade kit. A cover sheet referencing the appropriate replacement procedure for this upgrade will come in the plenum upgrade kit.

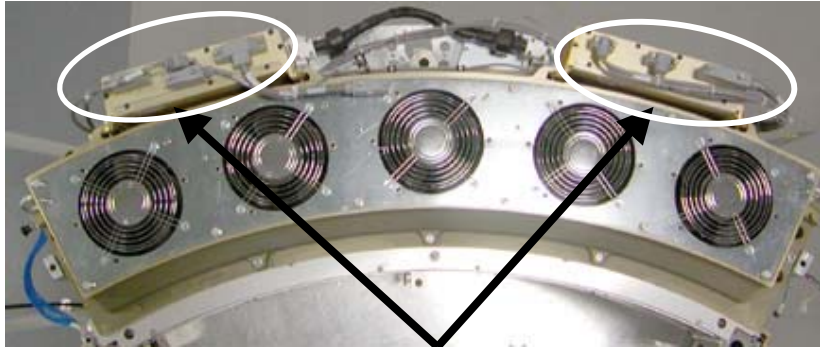
Attached below for reference is the updated Block diagrams for the new plenum. The next service document release will contain both old (2005 style) and new (2006 style) plenum block diagrams.

Steve Merritt
FCT Service Engineering

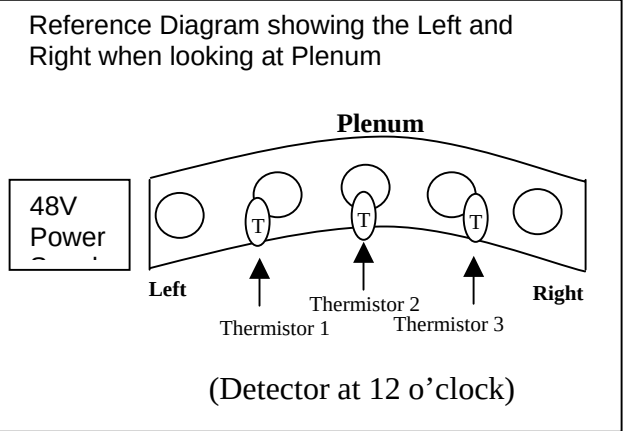
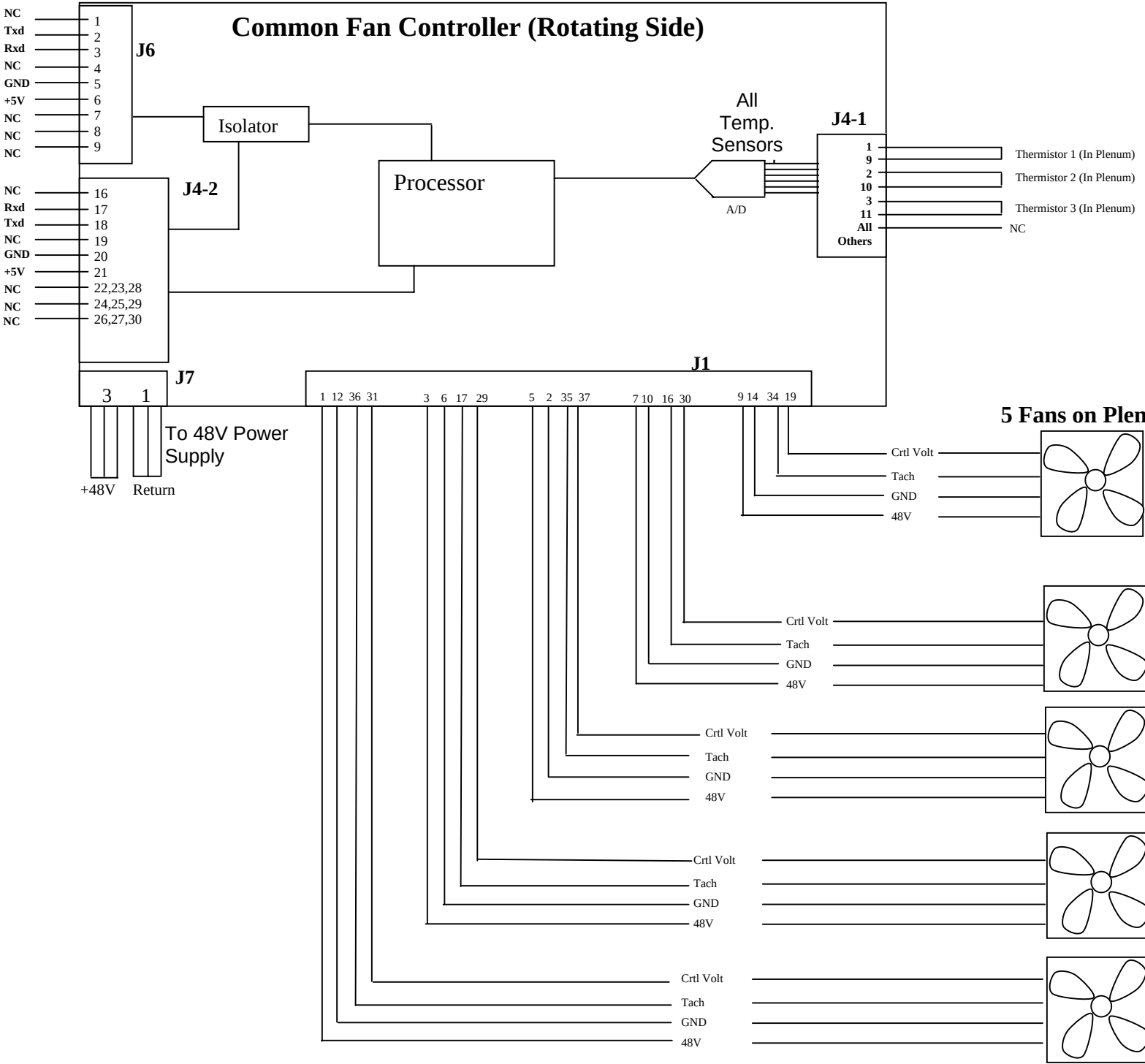
VCT Detector Plenum Fan Control (2006 Style)

RS232 to DHC
(Detector Heater
Controller on the
Plenum)

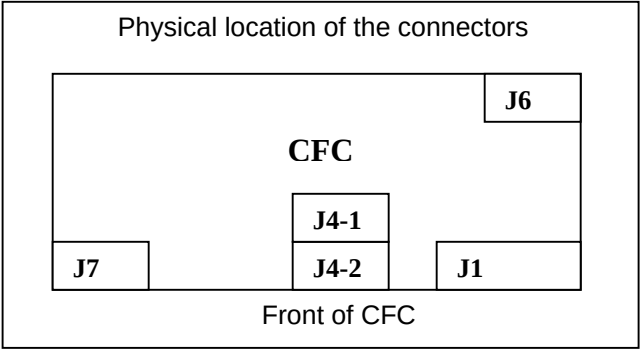
RS232 to
Backplane



2006 Style: Sub-D connector style for all CFC and DHC cables

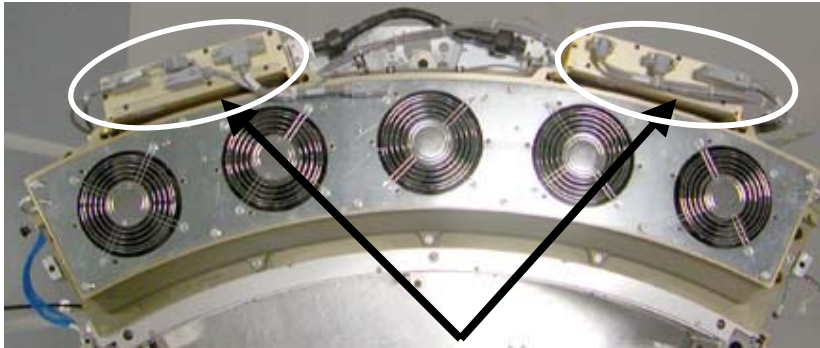


5 Fans on Plenum
Left Plenum Fan near 48V PS



Right Plenum Fan near ORP

VCT Detector Heater Control (2006 Style)



2006 Style: Sub-D connector style for all CFC and DHC cables

Rail Heater Resistances:

Low side left routing board (input side):
J1 Pin 1,3 is 50-62 ohm (DHC J1 Pins 9,10)
J1 Pin 2,4 is 20-24 ohm (DHC J1 Pins 11,13)

Low side left routing board (Output side):
J7 Pin 1,3 is about 27 ohm (Left Rear)
J6 Pin 1,3 is about 27 ohm (Left Front)
J7 Pin 2 to J6 Pin 2 is 20-24 ohm (Center)
Must have high side board connected to read center zones.

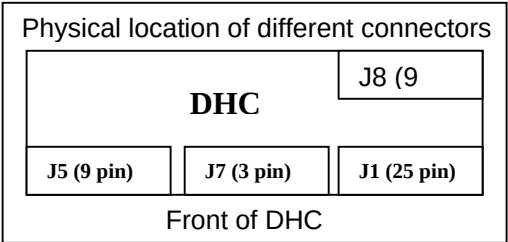
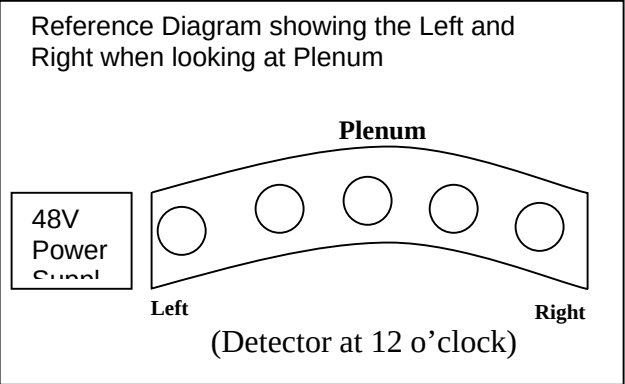
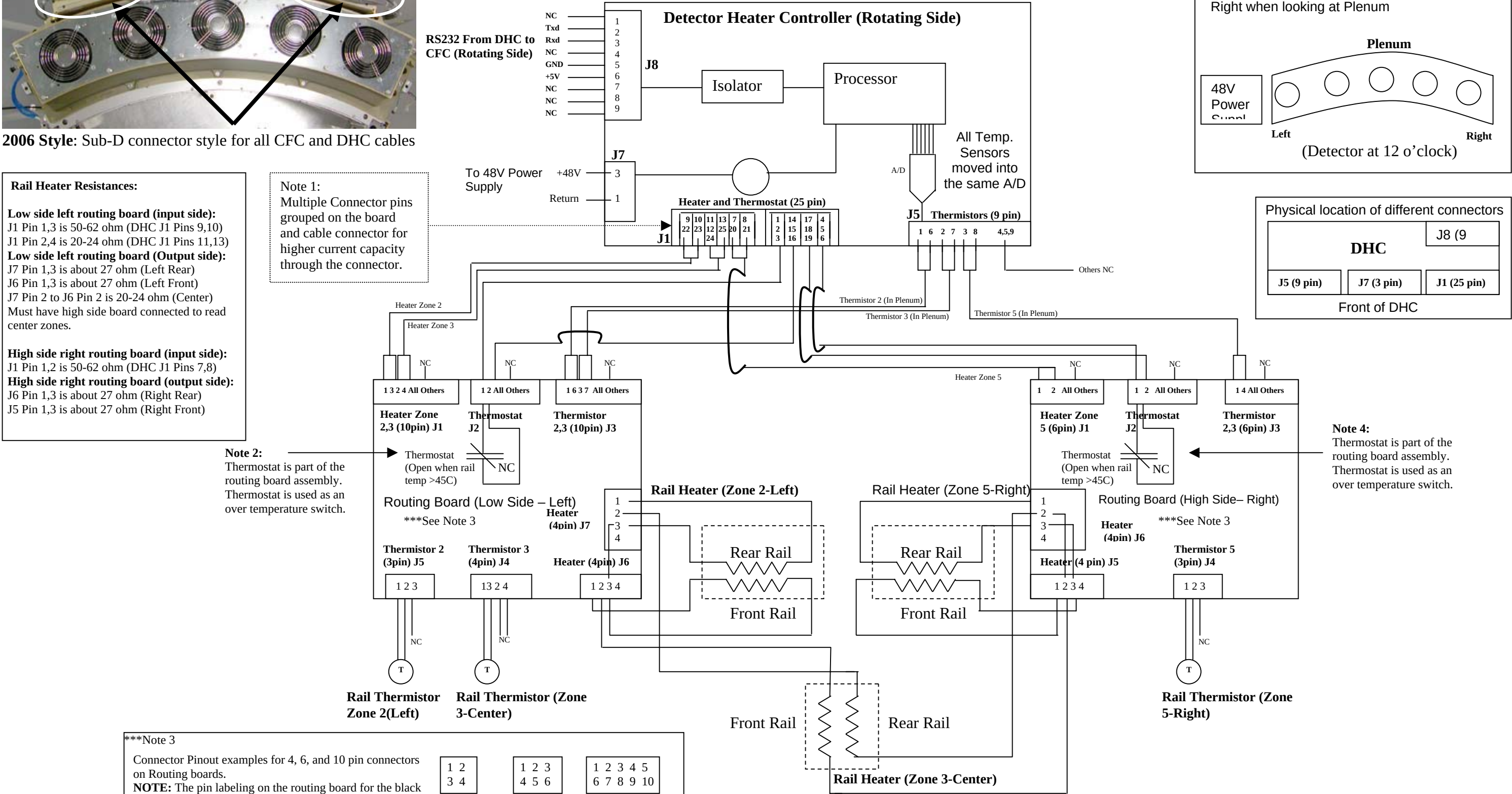
High side right routing board (input side):
J1 Pin 1,2 is 50-62 ohm (DHC J1 Pins 7,8)

High side right routing board (output side):
J6 Pin 1,3 is about 27 ohm (Right Rear)
J5 Pin 1,3 is about 27 ohm (Right Front)

Note 1:
Multiple Connector pins grouped on the board and cable connector for higher current capacity through the connector.

Note 2:
Thermostat is part of the routing board assembly. Thermostat is used as an over temperature switch.

Note 4:
Thermostat is part of the routing board assembly. Thermostat is used as an over temperature switch.



***Note 3

Connector Pinout examples for 4, 6, and 10 pin connectors on Routing boards.

NOTE: The pin labeling on the routing board for the black connectors is wrong. Use this diagram for reference. Pin 1 is correct on the board, the rest of the pins follow this order.

1 2 3 4	1 2 3 4 5 6	1 2 3 4 5 6 7 8 9 10
4 pin	6 pin	10 pin